

ABES ENGINEERING COLLEGE

Department of CSE- Data Science

50 Full Stack Development Mini-Project Ideas

For Second-Year Students

Node.js | Express | MongoDB | GraphQL | React | Docker | Deployment

Overview

This document lists 50 Full Stack Development project ideas for second-year students, organized by increasing difficulty. Each project maps to the concepts covered in the FSD lab syllabus — file handling, HTTP servers, Express routing, REST APIs, MongoDB CRUD and aggregation, GraphQL, Docker, and cloud deployment. Students may pick a project matching their comfort level or extend a Beginner idea into an Advanced one by adding auth, GraphQL, or containerization.

Beginner Level — CRUD & REST Fundamentals

Single-resource APIs, file/DB CRUD, basic routing

#	Project Title	Description
1	Student Record Manager	Express + MongoDB CRUD API with a simple React frontend to add/view/update/delete student records.
2	Personal Expense Tracker	REST API for expenses (category, amount, date) with a monthly summary endpoint.
3	Library Book Catalog	CRUD for books with borrow/return status and search by author/title.
4	Simple Blog API	Posts and comments as nested resources, with pagination on GET /posts.
5	Contact Book API	Store contacts with CRUD routes and search by name or phone number.
6	Movie Review Collector	Add movies and reviews, compute average rating per movie.
7	To-Do List API	Basic task CRUD with status (pending/done) — no authentication required.
8	Recipe Box	Store recipes with ingredients and steps; filter by cuisine or cook time.
9	Quote of the Day API	CRUD for quotes plus a random-quote endpoint.
10	Small Shop Inventory Tracker	Track stock items, quantities, and low-stock flags via REST routes.
11	Feedback Collection Form + API	Simple form posts feedback to an Express API storing it in MongoDB.
12	Product Wishlist Manager	Users add/remove products to a wishlist; list persisted in MongoDB.
13	Notes App API	Create, read, update, delete personal notes with timestamps.
14	Address Book with City Search	CRUD contacts with a query endpoint to filter by city.
15	Basic Polling App	Create polls, cast votes, and view live vote counts.

Intermediate Level — Auth, Relations & Real-Time

JWT auth, related collections, sockets, deployment basics

#	Project Title	Description
16	Task Manager with JWT Auth	Users log in and manage only their own tasks; protected routes via JWT middleware.
17	Employee Attendance Tracker	Mark daily attendance and generate monthly reports using MongoDB aggregation.
18	E-commerce Product Catalog with Cart	Products and categories with a session-based cart and checkout summary.

#	Project Title	Description
19	Event Registration Portal	Create events, register attendees with capacity limits, send confirmation via Nodemailer.
20	Recipe Sharing App	Users post recipes and like/favorite others' recipes; filter by cuisine.
21	Blogging Platform with Roles	Author and Reader roles; authors can publish, readers can only comment.
22	Online Quiz Application	Timed quizzes with auto-scoring and a results history per user.
23	Job Application Tracker	Personal tracker for job applications with status (applied/interview/offer).
24	Expense Splitting App	Mini Splitwise — split group expenses and calculate who owes whom.
25	Habit Tracker	Daily habit check-ins with streak calculation and progress charts.
26	Movie Seat Booking (Mock)	Select seats for a show and lock them for booking — no real payment.
27	Hostel Complaint / Grievance Portal	Students raise complaints, admins update status; track resolution time.
28	Mini CMS with Image Upload	Manage blog posts with image uploads stored via Multer.
29	Freelancer Portfolio Builder	CRUD for projects and skills, rendered as a public portfolio page.
30	Restaurant Menu & Order Management	Manage menu items and simulate order placement with order history.
31	Fitness Class Booking System	Book slots for fitness classes with capacity limits per session.
32	Personal Finance Dashboard	Express API + Chart.js frontend visualizing income/expense trends.
33	Multi-User Chat Room	Real-time chat using Socket.io with Express, supporting multiple rooms.

Advanced Level — GraphQL, Docker & Full Deployment

Multi-service stacks, containerization, cloud deployment

#	Project Title	Description
34	Student Management System (SMS)	React + Express + MongoDB, fully Dockerized and deployed on Render.
35	GraphQL Course Enrollment System	Students, Courses, and Enrollments as GraphQL types with resolvers over MongoDB.
36	Dockerized Real-Time Notes App	Node + MongoDB in Docker Compose with WebSocket sync across sessions.
37	Mini Job Portal	Employer and Candidate roles, job postings and applications, Dockerized.
38	Inventory Management with GraphQL	Query stock by category/supplier through a GraphQL API; low-stock alerts.
39	Hostel Room Allocation System	Allocate rooms based on availability; Dockerized end-to-end deployment.
40	Library System with GraphQL + Admin Panel	GraphQL API for catalog/borrowing with a React admin dashboard.
41	Online Learning Platform	Courses, lessons, and per-student progress tracking with a React UI.
42	E-commerce Backend with GraphQL	Products, orders, and users on GraphQL + MongoDB, containerized.
43	Real-Time Collaborative Whiteboard	Socket.io-based shared canvas, packaged with Docker.

#	Project Title	Description
44	Multi-Tenant Departmental Blog	One platform serving separate blogs per department (e.g., CSE-DS, CSE-AIML).
45	Smart Attendance Hook System	Attendance API that calls an external recognition service; Dockerized.
46	Ticket Booking with Seat Locking	Concurrency-safe seat locking using MongoDB transactions.
47	Fitness Tracker with GraphQL + Compose	Mongo + Node + React stack orchestrated via Docker Compose.
48	Campus Event Management Portal	End-to-end event lifecycle management, Dockerized and deployed on Render.
49	Online Voting System	Role-based access control for voters/admins, secured and Dockerized.
50	Full-Stack Placement Portal	Companies, students, and applications on React + Express + MongoDB + GraphQL, Dockerized and deployed on Render.